

The Complete Fractal Art Maker Program

Enter the following code into a new file and save it as [fractalArtMaker.py](#). This program relies on Python's built-in `turtle` module, so no JavaScript code is used for this chapter's project:

Python

```
import turtle, math

DRAW_FRACTAL = 1 # Set to 1 through 11 and run the program.

turtle.tracer(5000, 0) # Increase the first argument to speed up the drawing.
turtle.hideturtle()

def drawFilledSquare(size, depth):
    size = int(size)

    # Move to the top-right corner before drawing:
    turtle.penup()
    turtle.forward(size // 2)
    turtle.left(90)
    turtle.forward(size // 2)
    turtle.left(180)
    turtle.pendown()

    # Alternate between white and gray (with black border):
    if depth % 2 == 0:
        turtle.pencolor('black')
        turtle.fillcolor('white')
    else:
        turtle.pencolor('black')
        turtle.fillcolor('gray')

    # Draw a square:
    turtle.begin_fill()
    for i in range(4): # Draw four lines.
        turtle.forward(size)
        turtle.right(90)
    turtle.end_fill()
```